

ISO 2553 weld symbols chart: symbols and process numbers

1. Elementary symbols

The elementary symbol describes the joint preparation and weld shape. It is drawn on the reference line (the grey line in the table).

Symbol	Weld	Use
	Square butt (I)	Thin plates butted, no bevel
	Single-V butt	Both edges bevelled: butt weld on thicker plates
	Single-bevel butt	Only one edge bevelled: tee or corner joint with penetration
	Single-V with broad root face (Y)	V with a flat root face
	Single-U butt	Rounded-bottom groove, for heavy sections
	Single-J butt	Half-U: only one edge machined with a radius
	Fillet	The most common: tee, corner or lap joint
	Backing run	Run made on the back of a butt weld, after the main weld
	Plug / slot	Weld in a hole or slot in the upper part
	Spot	Resistance or fusion spot welds
	Seam	Overlapping spots forming a tight seam

For a weld on both sides, the symbol is drawn on each side of the reference line, without the dashed line: two Vs make an X, two triangles a double fillet.

2. Supplementary symbols

Symbol	Meaning	Effect
	Flat (flush)	Weld face must be flat, often after grinding
	Convex	Domed weld face

Symbol	Meaning	Effect
	Concave	Hollow face blending into the parts
	Weld all around	Circle at the arrow bend: the weld goes all around the part
	Field weld	Flag at the arrow bend: weld made on site, not in the shop

A backing strip is shown by M in a rectangle (permanent) or MR (removable), under the joint symbol.

3. Dimensions and process

No.	Process (ISO 4063)
111	Manual metal arc welding (stick)
121	Submerged arc welding with one wire electrode
131	MIG welding (solid wire, inert gas)
135	MAG welding (solid wire, active gas)
136	MAG welding with flux-cored wire
141	TIG welding (tungsten electrode, inert gas)

Check that the document's drafting standard is ISO (Document Properties → Drafting Standard) to get the reading described here. Hole and thread sizing for neighbouring parts is in the metric thread chart.

Reference values taken from the article <https://www.baouch.fr/en/blog/symboles-soudure-iso-2553/>, which explains the assumptions, formulas and edge cases. For critical use, check the current standard or the manufacturer's documentation. Free to print and share with credit to the source.